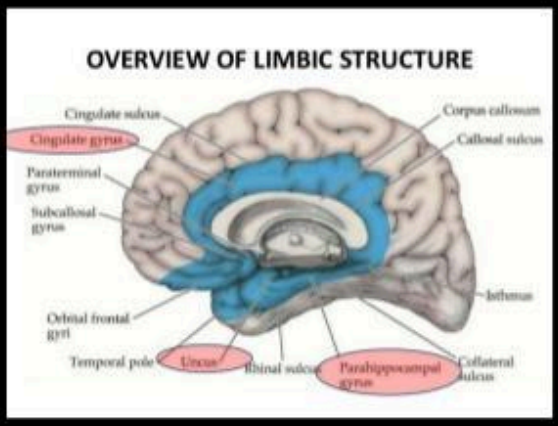
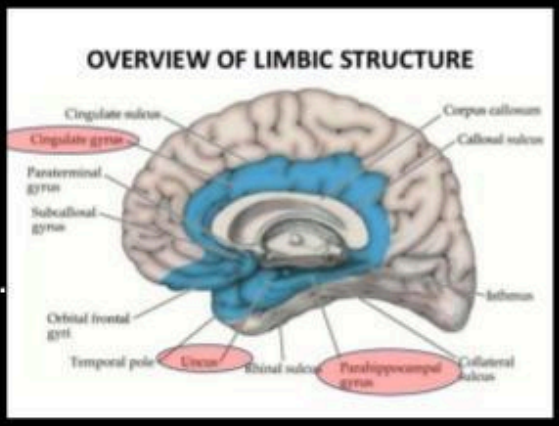
PHYSIOLOGICAL ANATOMY

- **Limbic cortex – Phylogenetically older part (allocortex)**
- **Composed of**
  - Orbitofrontal cortx
  - Subcallosal gyrus
  - Cingulate gyrus.
  - Parahippcampal gyrus.
  - Uncus.



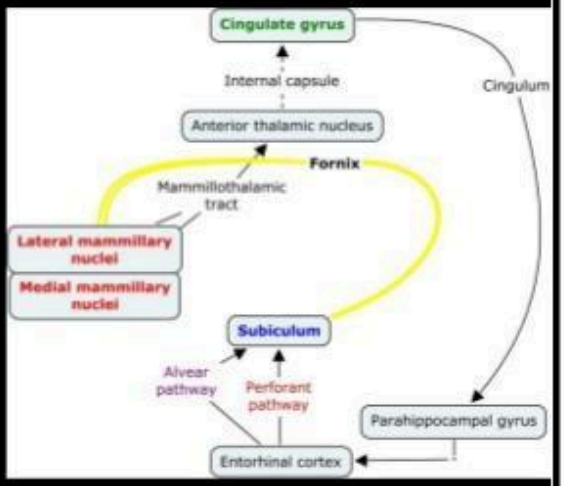
PHYSIOLOGICAL ANATOMY

- **Sub cortical structures includes –**
  - Hypothalamus
  - Septum
  - Paraolfactory area
  - Anterior nuclei of thalamus.
  - Amygdala
  - Portions of basal ganglia
  - Hippocampus.



CONNECTIONS

- **Papez circuit**
  - Closed circuit formed by connections between 4 nuclei.
- Responsible for resting EEG.
- For emotions & behaviour for preservation of species & individuals.

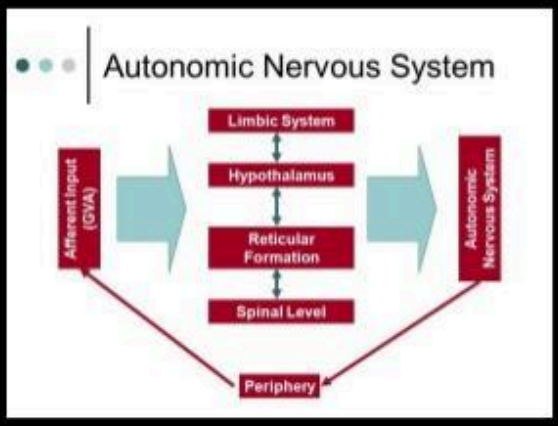


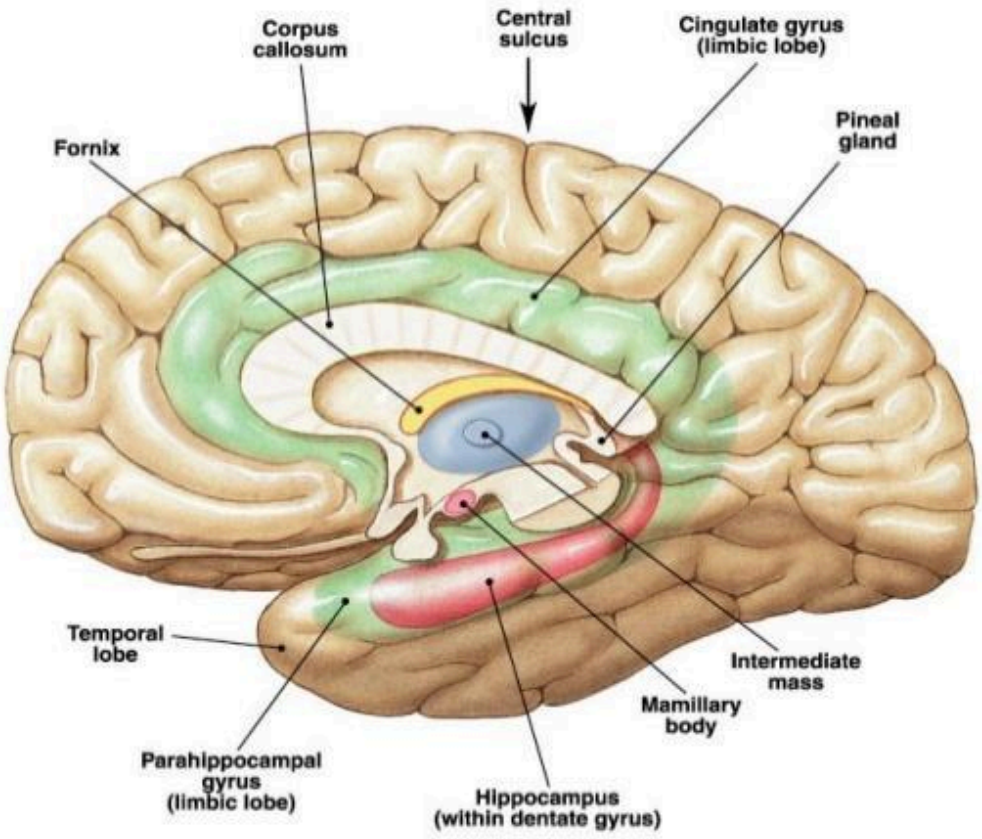
CHARACTERISTIC FEATURES OF CONNECTIONS.

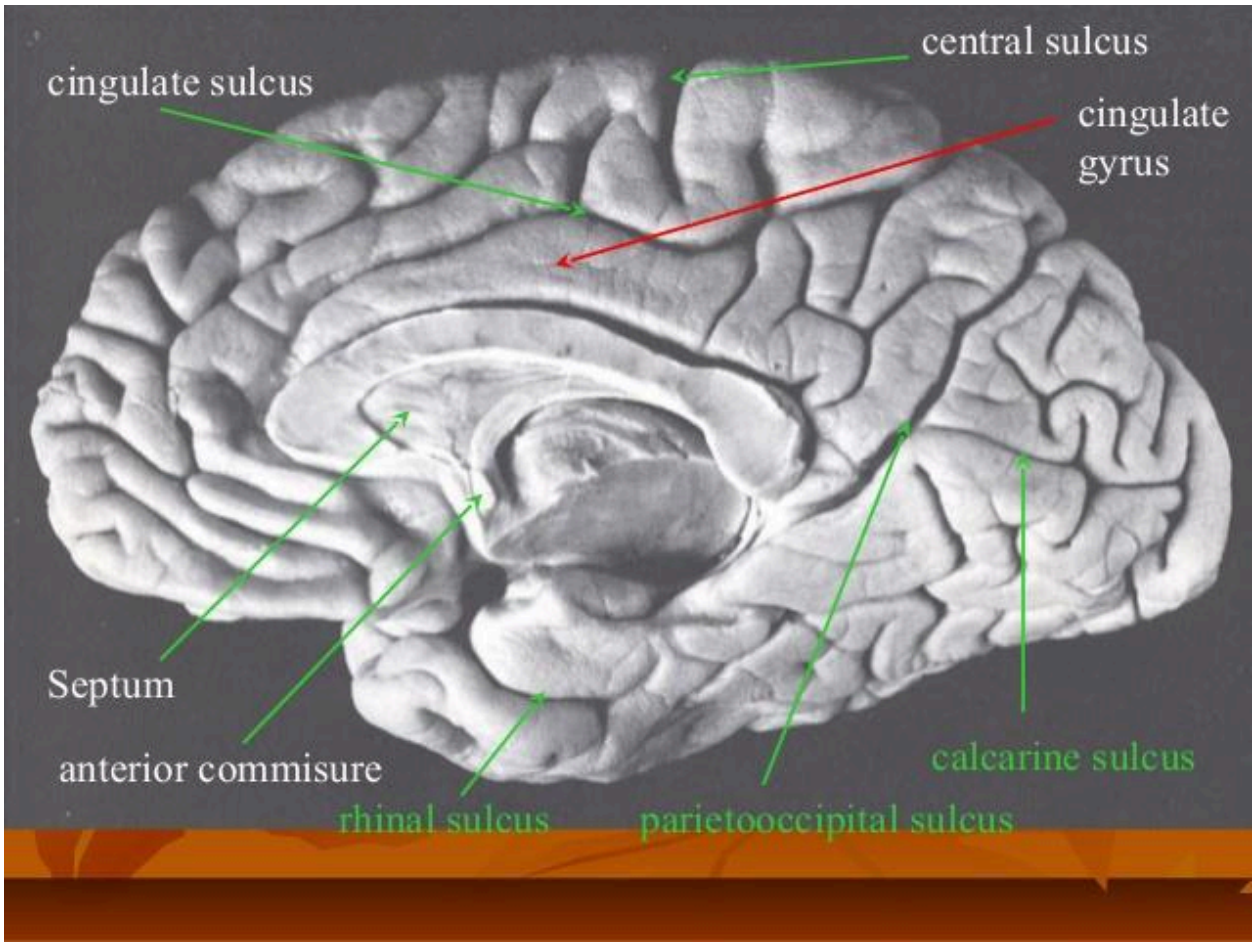
- Very little connections with Neo-cortex.
- So **Emotions** & **Instinctual behaviour** not under voluntary control.
- Prolonged **After discharge** – So emotional response continue long after end of stimuli which produces them.

AUTONOMIC FUNCTIONS.

- Stimulation of mainly – **Amygdala** through **Hypothalamus** causes Autonomic response.
- During emotional state mainly affect
  - Cardiovascular system
  - Respiratory system
  - GIT.

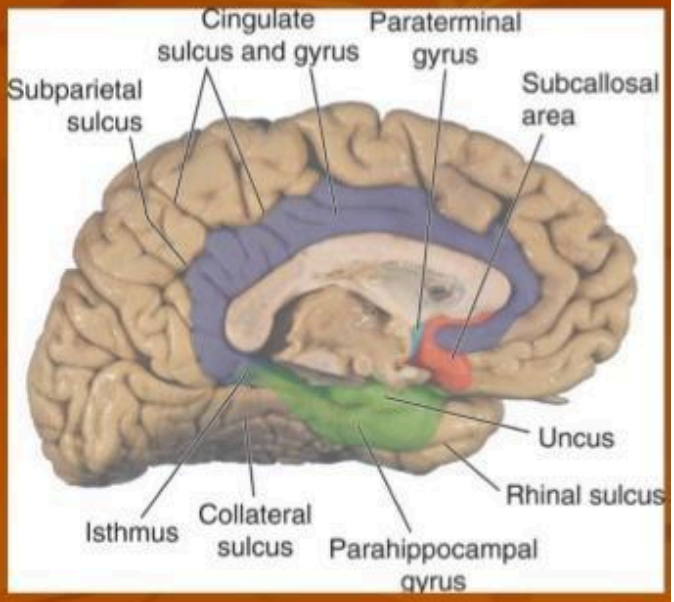



Corpus callosum
Central sulcus
Cingulate gyrus (limbic lobe)
Pineal gland
Fornix
Temporal lobe
Intermediate mass
Mamillary body
Parahippocampal gyrus (limbic lobe)
Hippocampus (within dentate gyrus)


central sulcus
cingulate sulcus
cingulate gyrus
Septum
anterior commisure
rhinal sulcus
parietooccipital sulcus
calcarine sulcus

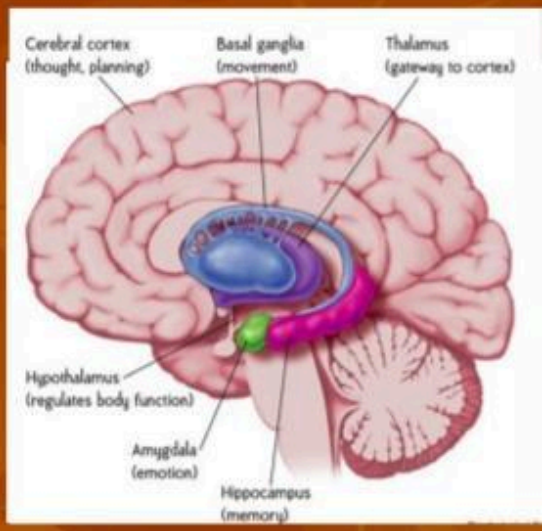
Parahippocampal gyrus

- Site: it lies on the inferior surface of each cerebral hemispher medial to the anterior part of the collateral sulcus
- Anteriorly continuous with the uncus
- Posteriorly continuous with the cingulate gyrus



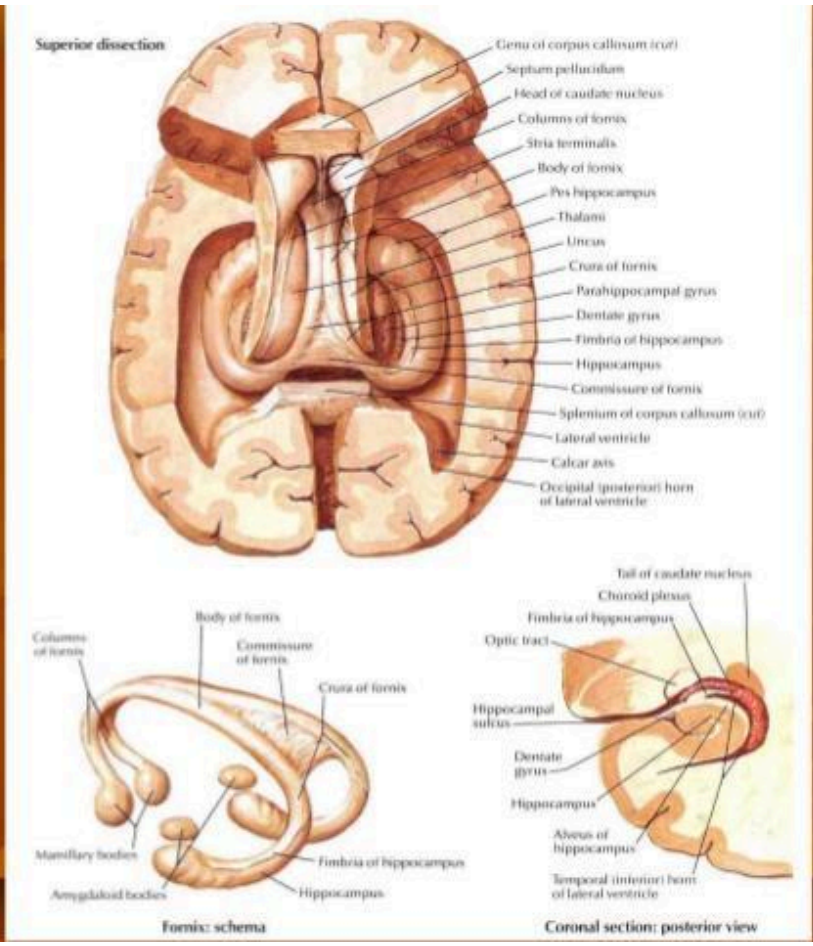
Amygdala

- Large nuclear group in temporal lobe. Almond in shape
- Relations: It is situated partly anterior and partly superior to the tip of the inferior horn of the lateral ventricle. It is fused with the tip of the tail of the caudate nucleus. The stria terminalis emerges from its posterior aspect
- Divisions
- Large basolateral region: Provides direct input to basal ganglia and motor system.
- Small corticomedial group of nuclei: Related to olfactory cortex.
- Medial and central nuclei: Connected to hypothalamus



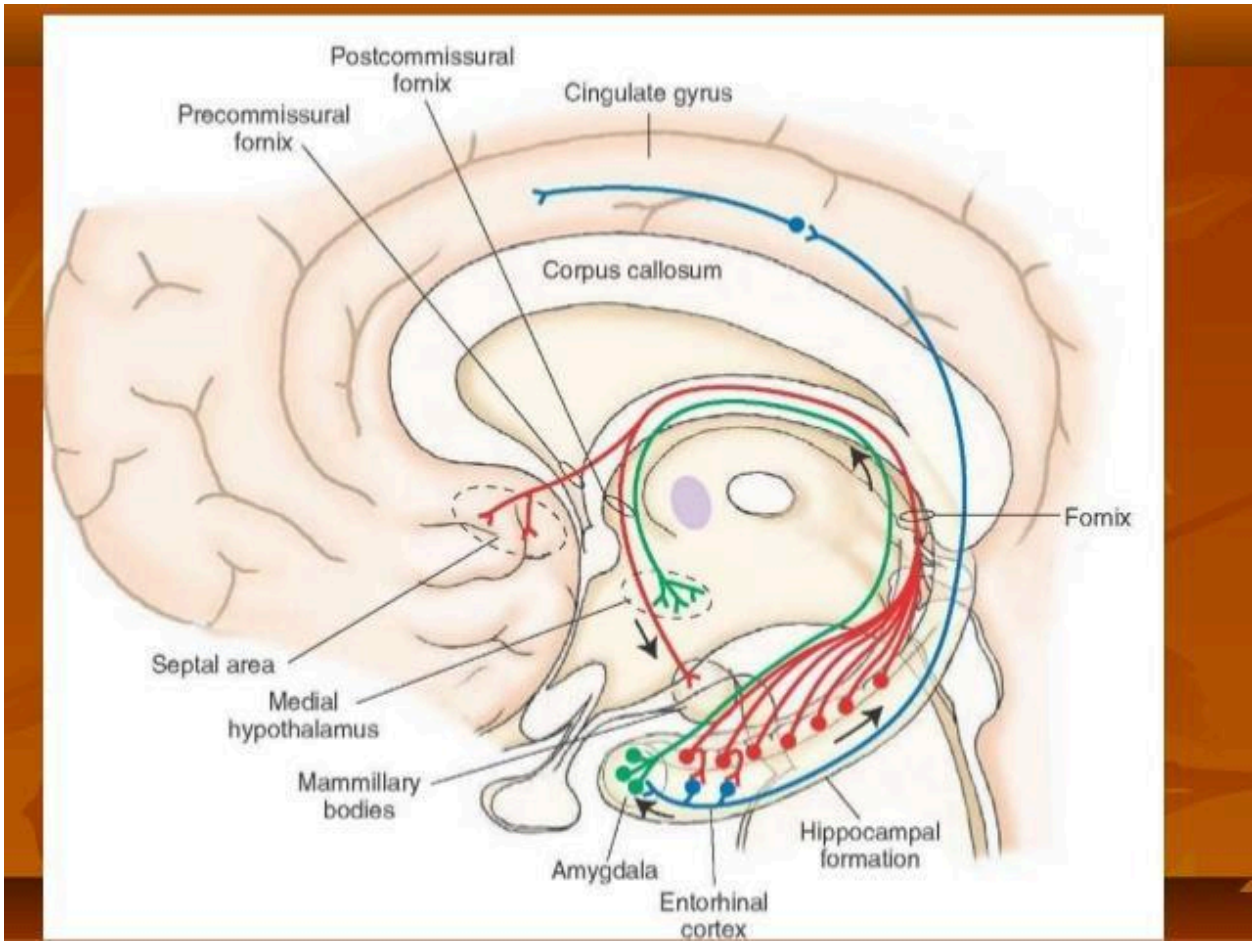
The hippocampal formation

- The hippocampal formation consists of hippocampus, dentate gyrus and subiculum
- The hippocampus is a curved elevation of gray matter and it is a part of the cerebral cortex enrolled in the floor of the inferior horn during the development. Its anterior end is expanded to form the pes hippocampus
- The hippocampus extends throughout the entire length of the floor of the inferior horn of the lateral ventricle
- In the coronal section the hippocampus resembles a sea horse
- The hippocampus terminates posteriorly beneath the splenium of the corpus callosum.


Superior dissection
Genu of corpus callosum (cut)
Septum pellucidum
Head of caudate nucleus
Columns of fornix
Stria terminalis
Body of fornix
Pes hippocampus
Thalamus
Uncus
Crura of fornix
Parahippocampal gyrus
Dentate gyrus
Fimbria of hippocampus
Hippocampus
Commissure of fornix
Splenium of corpus callosum (cut)
Lateral ventricle
Calcar avis
Occipital (posterior) horn of lateral ventricle
Body of fornix
Columns of fornix
Commissure of fornix
Crura of fornix
Mamillary bodies
Amygdaloid bodies
Fimbria of hippocampus
Hippocampus
Fornix: schema
Tail of caudate nucleus
Choroid plexus
Fimbria of hippocampus
Optic tract
Hippocampal sulcus
Dentate gyrus
Hippocampus
Alveus of hippocampus
Temporal (inferior) horn of lateral ventricle
Coronal section: posterior view

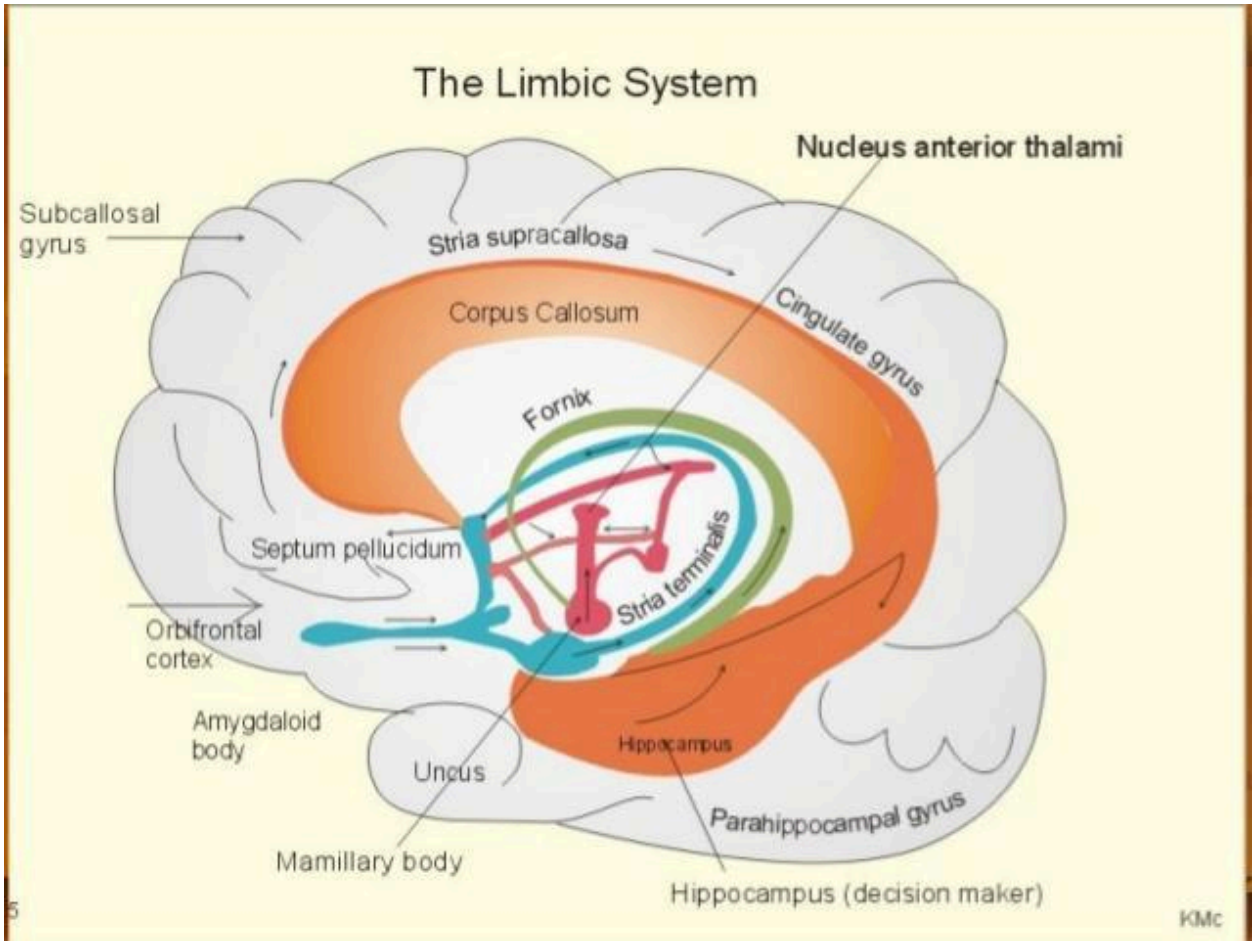
Afferent Connections of the Hippocampus

- Fibers from the cingulate gyrus pass to the hippocampus.
- Fibers from the septal nuclei (nuclei lying within the midline close to the anterior commissure) pass posterior in the fornix to the hippocampus.
- Fibers from one hippocampus pass across the midline to the opposite hippocampus in the commissure of the fornix.
- Fibers from the indusium griseum pass posteriorly in the longitudinal striae to the hippocampus.
- Fibers from the entorhinal area or olfactory-associated cortex pass to the hippocampus.
- Fibers arising from the dentate and parahippocampal gyri travel to the hippocampus.
- Fibers from the thalamus. the anterior nuclei and the midline complex, project to the hippocampal and parahippocampal areas.


Postcommissural fornix
Precommissural fornix
Cingulate gyrus
Corpus callosum
Fornix
Septal area
Medial hypothalamus
Mammillary bodies
Amygdala
Entorhinal cortex
Hippocampal formation

Efferent Connections of the Hippocampus

- The fibers within the fornix are distributed to the following regions:
- 1. Fibers pass posterior to the anterior commissure to enter the mammillary body, where they end in the medial nucleus.
- 2. Fibers pass posterior to the anterior commissure to end in the anterior nuclei of the thalamus.
- 3. Fibers pass posterior to the anterior commissure to enter the tegmentum of the midbrain.
- 4. Fibers pass anterior to the anterior commissure to end in the septal nuclei, the lateral preoptic area, and the anterior part of the hypothalamus.
- 5. Fibers join the stria medullaris thalami to reach the habenular nuclei.


The Limbic System
Nucleus anterior thalami
Subcallosal gyrus
Stria supracallosa
Corpus Callosum
Cingulate gyrus
Fornix
Septum pellucidum
Stria terminalis
Orbitfrontal cortex
Amygdaloid body
Hippocampus
Uncus
Parahippocampal gyrus
Mamillary body
Hippocampus (decision maker)
KMc

- The classic Papez circuit for emotions (1937) comprises of;
- hippocampus → fornix → mammillary body → mammillothalamic tract → anterior thalamic nuclear group → cingulate gyrus → parahippocampal gyrus and neural linkages back to the hippocampus